

SOCHI, Russian Federation

WMO: 371710

Lat: 43.450N

Lon: 39.957E

Elev: 89

StdP: 14.65

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 28.7	(c) 31.7	(d) 19.7	(e) 14.9	(f) 34.0	(g) 23.1	(h) 17.4	(i) 36.7	(j) 22.2	(k) 48.1	(l) 18.8	(m) 46.8	(n) 7.1	(o) 70	(p) 0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 13.8	(c) 87.7	(d) 74.9	(e) 85.6	(f) 74.7	(g) 83.8	(h) 73.8	(i) 78.1	(j) 84.2	(k) 76.6	(l) 82.9	(m) 75.2	(n) 81.7	(o) 7.7	(p) 290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.7	135.1	82.6	74.8	130.6	81.7	73.2	123.7	80.2	41.6	84.5	40.2	82.9	38.9	81.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 16.7	(b) 14.7	(c) 13.3	(e) 24.3	(f) 92.7	(g) 3.5	(h) 3.5	(i) 21.8	(j) 95.2	(k) 19.7	(l) 97.3	(m) 17.7	(n) 99.3	(o) 15.2	(p) 101.8		
(4) 16.7	(5) 14.7	(6) 13.3	(7) 22.8	(8) 80.0	(9) 3.3	(10) 1.4	(11) 20.5	(12) 81.1	(13) 18.5	(14) 81.9	(15) 16.6	(16) 82.7	(17) 14.2	(18) 83.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.6	43.7	44.1	47.6	54.3	62.2	69.8	74.7	76.1	69.3	61.4	52.7	46.4
	DBStd	12.77	5.81	6.33	6.61	6.40	5.90	4.82	3.98	3.83	4.86	5.60	5.95	6.19
	HDD50	731	209	185	127	26	0	0	0	0	0	2	39	143
	HDD65	3349	661	586	540	325	125	13	0	0	18	137	368	576
	CDD50	3877	13	19	53	156	380	593	767	808	579	355	122	32
	CDD65	1020	0	0	1	5	39	156	302	343	147	26	1	0
	CDH74	7249	0	0	7	36	201	828	2381	2894	795	105	2	0
	CDH80	1615	0	0	1	4	37	129	551	770	114	9	0	0
	WSAvg	6.5	7.4	7.0	6.8	6.0	5.6	6.2	6.2	6.4	6.5	6.3	6.8	7.3
	PrecAvg	62.3	6.4	5.4	5.4	4.2	4.4	3.9	4.1	3.7	5.6	7.0	6.3	6.1
(15) Precipitation	PrecMax	81.7	10.0	10.1	11.8	7.6	12.4	9.6	8.5	9.4	18.4	17.6	15.6	12.7
	PrecMin	47.5	1.6	1.3	2.0	1.5	0.8	0.7	1.3	0.2	0.7	1.4	0.6	2.0
	PrecStd	8.6	2.2	2.2	2.0	1.6	2.7	2.0	2.2	2.3	3.5	3.9	3.7	2.4
	PrecStd	8.6	2.2	2.2	2.0	1.6	2.7	2.0	2.2	2.3	3.5	3.9	3.7	2.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.4	65.4	72.8	78.4	84.3	86.4	91.2	90.0	86.3	80.7	72.0	66.6
		MCWB	49.7	50.5	54.6	59.0	65.4	71.3	73.5	74.5	70.9	65.4	56.6	51.3
	2%	DB	58.9	60.9	66.3	73.4	79.1	83.4	87.7	87.8	83.3	76.9	68.2	62.2
		MCWB	47.8	48.5	51.9	57.7	64.7	71.9	75.6	76.0	70.9	65.6	56.0	51.1
	5%	DB	55.3	57.3	62.4	68.6	75.3	80.8	84.6	85.9	80.8	73.7	64.8	58.6
		MCWB	46.7	47.2	50.7	56.3	64.0	71.5	75.0	75.7	70.6	64.2	55.6	50.4
	10%	DB	52.1	53.9	57.5	64.8	72.0	78.6	82.7	84.2	78.7	71.2	62.2	55.5
		MCWB	46.2	46.5	49.2	55.0	63.8	70.6	74.3	75.0	69.7	63.2	54.8	49.0
	0.4%	WB	51.5	52.7	56.2	61.4	69.5	76.1	79.5	79.9	76.0	69.5	60.3	55.7
		MCDB	57.4	61.1	67.2	71.7	76.1	82.2	85.5	86.1	82.9	76.8	66.0	60.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	49.8	50.4	53.5	59.3	67.4	74.2	78.0	78.4	73.2	66.9	58.6	53.0
		MCDB	54.7	56.4	62.9	68.4	74.2	80.1	83.9	84.6	79.7	73.7	64.1	58.3
	5%	WB	48.4	48.7	51.8	57.7	65.8	72.7	76.4	77.2	71.7	65.3	57.0	51.2
		MCDB	53.2	54.5	59.1	66.2	72.5	78.8	82.5	83.6	78.5	71.9	62.9	56.6
	10%	WB	46.9	47.4	50.3	56.1	64.5	71.3	75.0	75.8	70.1	63.6	55.3	49.7
		MCDB	51.4	52.9	56.0	63.3	71.1	77.3	81.4	82.3	77.1	69.7	61.2	54.7
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	12.9	13.0	14.4	13.8	13.4	13.5	13.8	14.9	14.7	15.2	13.2
		MCDBR	16.1	17.8	20.3	21.3	19.2	15.4	15.1	14.7	16.5	18.0	18.7	16.9
		MCWBR	9.3	9.4	9.9	10.2	9.8	9.1	9.0	8.9	10.1	11.0	11.3	10.0
	5% WB	MCDBR	13.4	14.6	16.8	17.9	15.5	13.5	13.6	13.4	14.9	15.8	16.4	14.4
		MCWBR	8.7	8.6	9.1	9.4	8.8	8.7	9.0	9.0	9.8	10.4	10.9	9.4
(40) Clear-Sky Solar Irradiance	taub		0.354	0.388	0.440	0.485	0.453	0.445	0.463	0.466	0.418	0.402	0.362	0.347
	taud		2.279	2.169	2.055	1.995	2.193	2.277	2.247	2.232	2.343	2.326	2.366	2.327
	Ebn at Noon		238	248	249	248	262	264	257	252	256	243	235	229
	Edh at Noon		28	37	47	54	45	42	43	42	35	31	26	24
	RadAvg		439	694	989	1410	1755	1992	2030	1789	1372	917	599	406
(45) All-Sky Solar Radiation	RadStd		60	90	85	177	192	137	118	113	150	94	72	55

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+274	N/A
(47) Regional (5 neighbors)	+0.97	N/A	N/A	N/A	N/A	N/A	N/A	-220	+212	N/A

Nomenclature: See separate page